# Chenxu Liu 刘宸旭

Undergraduate in Physics — Shandong University, Qingdao

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## EDUCATION

### B.Sc. in Physics — Shandong University, Qingdao

2024 – 2028

Institute of Frontier and Interdisciplinary Science

Coursework: quantum mechanics, electrodynamics, statistical physics, particle physics, computational methods. Senior thesis on TMD factorization planned for 2027–28.

## RESEARCH INTERESTS

Transverse-momentum dependent distributions and nucleon spin structure; SIDIS and Drell–Yan phenomenology; small weight-tied sequence models and the optimizers that suit them.

## PUBLICATIONS & PREPRINTS

- C. Liu, "Higher-twist corrections to polarized TMDs from Gegenbauer moments," manuscript in preparation, The European Physical Journal Plus, 2026.
- C. Liu, "Depth-aware optimizers for weight-tied feed-forward stacks," preprint, arXiv submission planned for November 2026.

## RESEARCH EXPERIENCE

### Undergraduate researcher — Particle Theory Group, SDU

2025 – present

- Built a Gaussian model with width$^2$ = $z^2\langle k_T^2\rangle$ + $\langle p_T^2\rangle$ to extract $\langle k_T^2\rangle$ from published SIDIS multiplicities, then cross-checked the result against JLab 12 GeV data from CLAS12 and Hall C.
- Twice-weekly reading group on spin physics; notes on twist-3 contributions and Gegenbauer moments circulated to the group.

## TEACHING & GRADING

### Grader and teaching assistant — Zhixin Education

2024 – present

- Graded national-semifinal physics olympiad (CPhO) scripts against a written rubric.
- Designed a rubric-driven grading pipeline with LLM assistance; it now processes a full exam batch end to end, with every score traceable to a rubric item.

## SELECTED PROJECTS

- **Loop_FFN.** Shared-weight loop FFN (SwiGLU) vs. stacked FFN on Higgs, depth sweep {1, 2, 4, 8, 16}; +0.43 pp at depth 4 ($p < 0.05$). Code, logs and reproduction scripts released.
- **exam-RAG.** Hybrid retrieval over 46 physics textbooks (ChromaDB) wired into a production grading pipeline, so that every judgement cites a source page.

## TECHNICAL SKILLS

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|---|---|
| LANGUAGES | Python (primary), C++, CUDA, Bash |
| SCIENTIFIC | PyTorch, ROOT, Geant4, Mathematica |
| METHODS | TMD model fitting, retrieval-augmented pipelines, ChromaDB hybrid search, LaTeX and TikZ |
| TOOLING | Git, HPC clusters and batch jobs, reproducible builds |

## LANGUAGES

Chinese (native) · English (working proficiency; IELTS in preparation, target band 7.0).